Splunk fundamentals

What is splunk and why do we need it

Splunk is a tool that is used to collect, search, monitor, analyze and visualize machine generated data.

Machine data :-

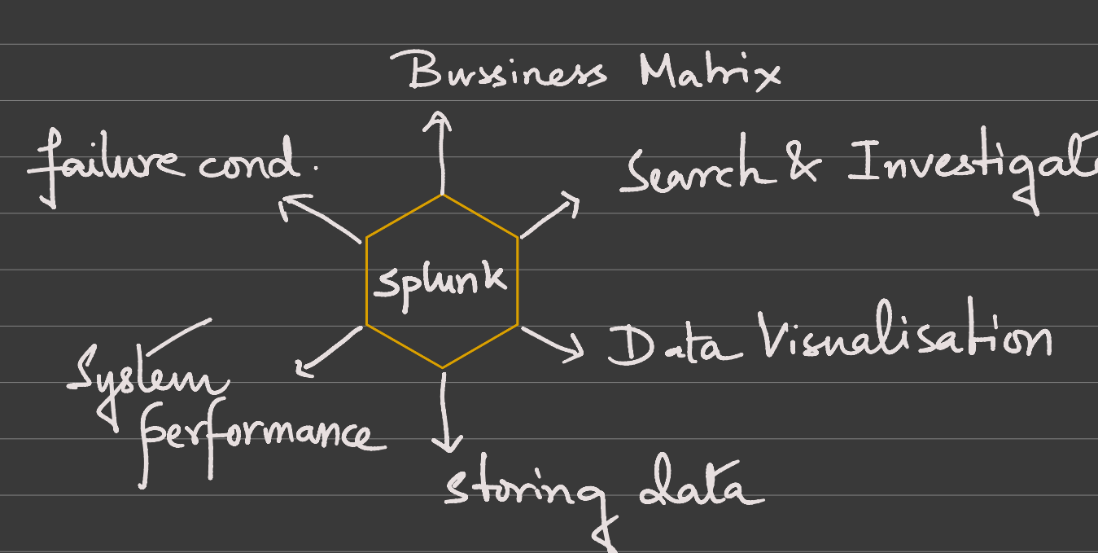
logs generated by Application, servers, databases, etc.

why → logs are complex to understand, unstructured in format, difficult to analyze.

Machine data —Splunk→ Human readable data

Components of Splunk:-

- forwarders → Collect data
  ↳ fwd 2th Splunk Instance.
- Indexers → Used as data Storage
- Search heads → Analyze, Visualize report.


Bussiness Matrix
failure cond.
Search & Investigate
splunk
System performance
Data Visualisation
storing data

Data Indexes ⟶ Users.
↳ Search processing languages

Why use Splunk as devops Engineer

Use case 1: Troubleshooting from server logs

Use case 2: Monitoring Infra / Appl. behaviour

3: Error detection.

4: Incident Investigation by analyzing events.

5: performance analysis.

6. Alerting when something goes wrong

7. Security → Investigate Suspicious activity

Data Sources → forwarder → Indexer (helps in storing data and indexes it) → Search head → Search report dashboard

→ Reports, Alerts, Visualization.

Input: (Collection) Splunk ingests raw data streams from servers, network devices, API logs, or log files.

parsing: System breaks data → Indv. Events, Timestamps, Metadata.

Indexing: parsed events compressed & written to disk as searchable indexes

Searching: Using query on indexed data to build dashboards.

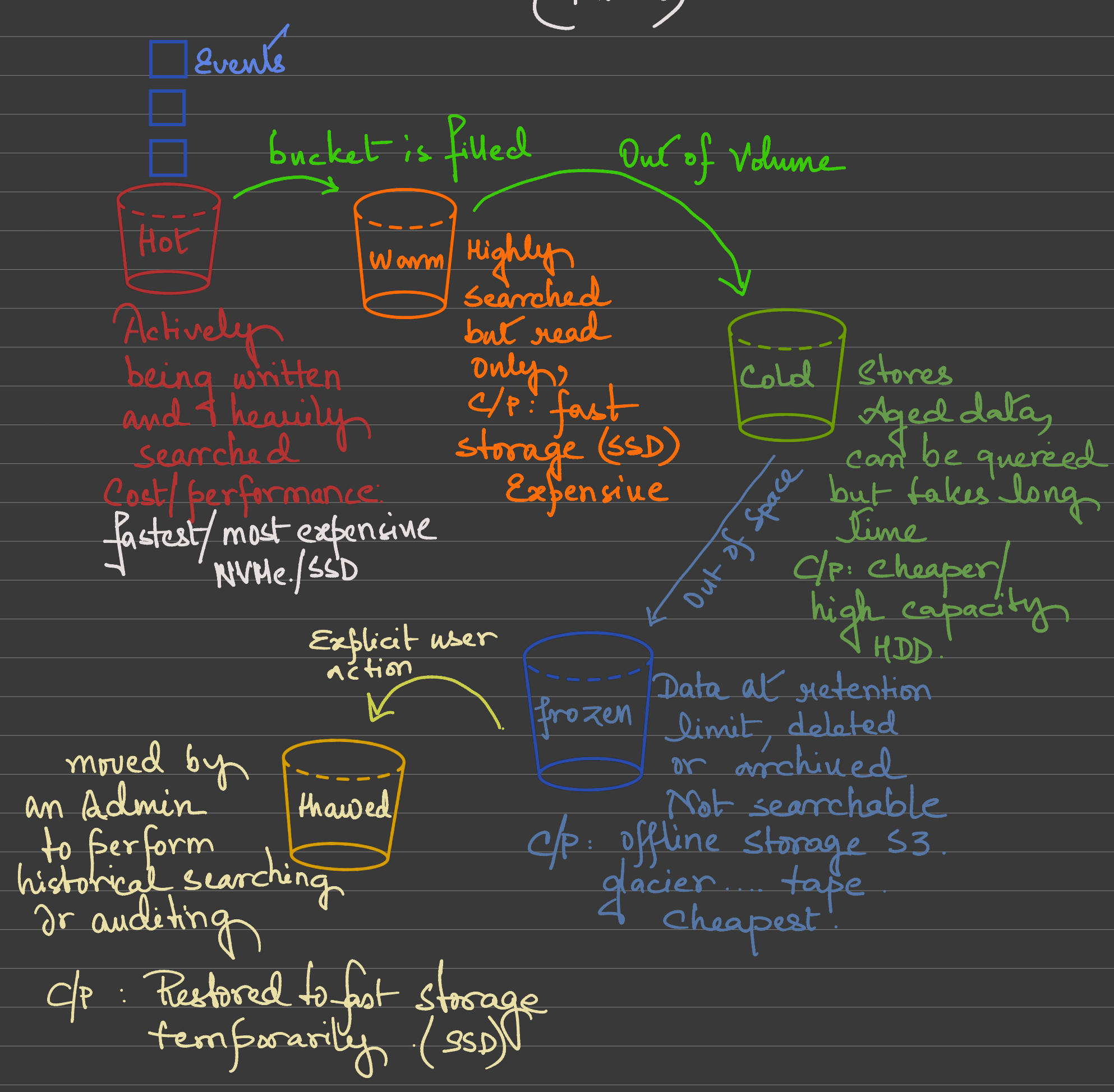
Data Onboarding & Parsing

'Holy Trinity' of parsing

- input.conf (Tells Splunk where to listen for data) ↳ Source
- props.conf (Defines how to handle data format) ↳ configure rules
- transforms.conf (Handles complex data routing and masking)
  ↳ helps to drop useless events, sensitive info.

Data lifecycle: (Buckets)

Indexers —flat files→ directories called (Bucket)


Events
bucket is filled
Out of Volume
Hot
Actively being written and heavily searched
Cost/performance:
fastest/most expensive NVMe/SSD
Warm
Highly searched but read only; C/P: fast storage (SSD) Expensive
Cold
Stores Aged data, can be quereed but takes long time
C/P: cheaper/ high capacity HDD.
Out of Space
frozen
Data at retention limit, deleted or archived
Not searchable
C/P: offline storage S3. glacier.... tape cheapest.
Explicit user action
Thawed
moved by an Admin to perform historical searching or auditing
C/P : Restored to fast storage temporarily (SSD)

Critical tool for DevOps.

- troubleshooting
- Security Monitoring
- Operational Intelligence

Splunk Architecture

Splunk pipeline relies on 3 main components.....

forwarders: Collects raw data from endpoints

(UF) universal forwarder → lightweight agent that collects data and ships data.

(HF) heavy forwarder → can parse and route data before sending it.

Indexer: Receives data, parses it → Events, writes it to disk (indexes) for storage.

Search head: Web UI where users write queries. It sends search req → Indexers results ↳ merges

Devops troubleshooting workflow

Incident Occurs. ↳ Generates an Event → Splunk → Investigate using time & correlation → Cause.

1. Identify the Spike:

timechart —obs.→ when errors started.

Eg. index = web status = 5* | timechart count span = 5m

2. Isolate Source: Group / Aggregate errors ↳ check Server, Container, Endpoint causing error.

Eg. index = web status = 5* | stats count by host, url | sort -count

3. Correlate by time:

pinpoint failing Endpoint / Specific timestamps
↳ Search for host / transaction_Id → errors ← to find out root cause of underlying

Easy breakdown of Code:

index = web. tells Splunk to Search inside bucket named "web"

status = 5* look for event where status code begins with 5* ↳ wildcard (501, 502, ...)

stats count by → group by specific category

timestamp count span = 5m → groups data into chunks of time

Search Processing language (SPL)

SPL is the language used to search and analyze splunk data.

Pipe character: (|) ⇝ pass the result of one command to the next command.

Essential Commands:

Base: index & _time

fields: keep or remove field to speed up search.

stats: Calculates aggregate statistics.

timechart: Creates timeseries req. _time.

eval : Calculates or creates new fields.

dedup: Remove duplicate events based on a field.

table : formats data into clean grid (Results)

How is Data Indexed.

field:	Desc.
_time	timestamp of the event (imp) } extracted raw Log
host	name or IP of machine → generated data.
source	file path script } → data came from. port
sourcetype	data format
index	logical bucket where data is stored on disk.

Data Ingestion: (How logs enter Splunk ecosystem)

- ☆ Universal forwarders → Installed Application / Web Servers to monitor local log files. { stream → Splunk.
- HTTP Event Collector (HEC)
  token —fast→ Endpoint —allows→ Docker Container, Serverless func
  ↳ JSON Logs ⇝ POST ⇝ Splunk
  No need of local Agent
- Syslog: Network device ⇝ UDP/TCP ⇝ Central Syslog Server ← (UF) ☆